AI for Scientific Writing — Session 1 handout

Glossary · acronyms · ten habits · AI-use log · disclosure — Semana Tec, Tec de Monterrey, Oct 2026

The rule for the week: *the model assists; you are the author.* You may use AI for any step you can explain, check and defend. Everything you submit — every sentence, number and reference — is your responsibility.

Course site — slides, handouts, schedule: `lrodriguez15.github.io/ai-scientific-writing.html`

Allowed tools in this course: TECgpt, Agent Studio, Gemini and Copilot (institutional); Zotero and Mendeley for references. AI is **not a reliable source**: validate what it tells you in other reliable sources.

The map

AI ⊃ **machine learning** ⊃ **deep learning** ⊃ **generative AI** ⊃ **large language models (LLMs)**. **Agentic AI** = an LLM + tools (search, code, browser, files) + a loop: goal → plan → act → observe → check. An LLM writes by predicting the next **token**. It optimises *plausible*, not *true* — which is why it can **hallucinate**: fluent, confident and wrong.

Glossary — 32 terms

Term	Plain meaning
Artificial intelligence (AI)	Systems that do tasks normally needing human intelligence.
Machine learning (ML)	AI that learns patterns from data instead of hand-written rules.
Deep learning	ML with many-layer neural networks.
Neural network	A model made of layers of weighted connections, trained on examples.
Generative AI (GenAI)	Models that create new text, images, code or audio.
Large language model (LLM)	GenAI for text, trained to predict the next token.
Foundation model	A large general-purpose model adapted to many tasks.
Transformer	The neural-network architecture behind today’s LLMs (attention mechanism).
Token	The unit an LLM reads and writes — a word piece (~¾ of a word in English).
Parameters	The learned numbers inside a model; more parameters ≈ more capacity.
Training / pre-training	Learning from a huge dataset before any specific use.
Fine-tuning	Extra training on a smaller, specific dataset.

Term	Plain meaning
RLHF	Reinforcement learning from human feedback — tuning a model toward answers people prefer.
Prompt	The input you give the model.
System prompt	Hidden or set-up instructions that shape every answer (role, rules).
Prompt engineering	Writing prompts that reliably get useful output.
Zero-shot / few-shot	Asking with no examples / with a few worked examples in the prompt.
Chain of thought / reasoning model	Getting the model to work step by step; models trained to “think” before answering.
Context window	How much text the model can consider at once (prompt + documents + reply).
Knowledge cutoff	The date after which the model’s training data stops.
Temperature	Randomness of the output: low = focused and repeatable, high = varied and creative.
Top-p (nucleus sampling)	Another randomness control: sample only from the most likely tokens.
Hallucination	Fluent, confident output that is false or invented (e.g. fake citations).
Grounding	Tying answers to provided sources so they can be checked.
RAG (retrieval-augmented generation)	The model first retrieves relevant documents, then answers from them.
Embeddings	Numbers that represent meaning, used to find similar texts (search, RAG).
Multimodal	Handles more than text: images, audio, video.
Agent / agentic AI	An LLM that plans and uses tools to take multi-step actions toward a goal.
Tool use / function calling	The model calling external tools (search, code, calculator, APIs).
Prompt injection	Malicious instructions hidden in content the AI reads (web page, PDF, email).
Bias	Systematic skew from training data or design that produces unfair or wrong outputs.
Guardrails / alignment	Measures that keep a model’s behaviour safe and within intended limits.

Acronyms

Acronym	Stands for
AI	artificial intelligence
ML	machine learning
DL	deep learning
GenAI	generative artificial intelligence
LLM	large language model
GPT	generative pre-trained transformer (the models behind ChatGPT)
RAG	retrieval-augmented generation
RLHF	reinforcement learning from human feedback
API	application programming interface — how programs and agents call a service
IEEE	Institute of Electrical and Electronics Engineers — publishes engineering papers and templates
IMRaD	Introduction, Methods, Results and Discussion — the usual shape of a paper

Acronym	Stands for
DOI	digital object identifier — a paper's permanent ID (doi.org/10.…)
APA	American Psychological Association — a citation style (IEEE uses numbered references [1])

Writing rule #1: spell out every acronym the first time — *large language model (LLM)* — then use the short form.

Six parts of a good prompt

Role (who it acts as) · **Task** (one clear verb) · **Context** (topic, audience, your material) · **Format** (length, sections) · **Examples** (one or two) · **Constraints** (what not to do; cite sources; say "I don't know"). Then iterate: one change at a time.

Ten habits that make AI useful

1. **Give context:** who you are, the goal, the audience, the format.
2. **Give it your material** (notes, data, papers) instead of asking it to recall facts.
3. **Ask for structure and critique first,** prose last.
4. **Iterate:** one change per prompt; keep what works.
5. **Verify** every fact, number and reference against the primary source.
6. **Protect data:** nothing confidential or unpublished in a tool whose terms you haven't checked.
7. **Ask for uncertainty:** "what are you least sure about?" and "where does this come from?"
8. **Keep a log** of the prompts and outputs that shaped your work.
9. **Rewrite in your own words** — if you can't explain a sentence, it isn't yours yet.
10. **Declare your use** — Tec's guidelines and every major publisher require it.

AI-use log (one row per AI interaction that shaped your paper)

#	Date	Tool & version	Section	What I asked (prompt, short)	What it gave	What I kept / changed / rejected	How I verified
1							

AI disclosure (IEEE style, in the Acknowledgments of your paper)

> *The authors used [tool, version] to [purpose: e.g. suggest the structure of Section II, review the draft for clarity, generate plotting code for Fig. 2]. AI-assisted content appears in [sections]. All AI output was reviewed, verified against the cited sources and our data, and edited by the authors, who take full responsibility for the content of this paper.*